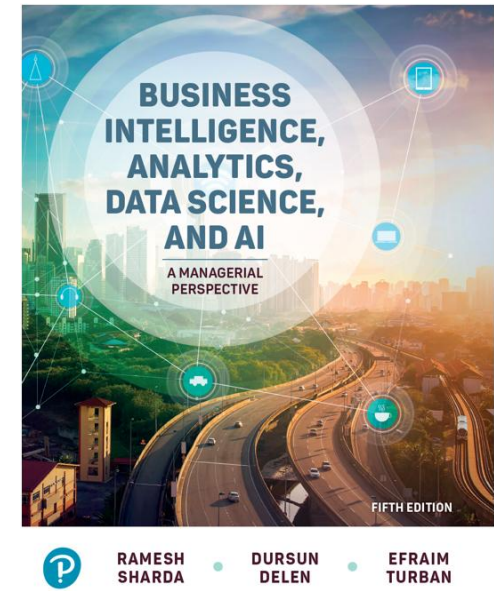


Chapter 11

Ethical, Privacy and Managerial Considerations in Analytics



Learning Objectives (1 of 2)

- 11.1 Describe the major implementation issues of intelligent technologies
- 11.2 Discuss legal, privacy, and ethical issues
- 11.3 Discuss the importance of ethical/responsible use of intelligent technologies
- 11.4 Understand the deployment issues of intelligent systems
- 11.5 Describe the major impacts on organizations and society

Learning Objectives (2 of 2)

- 11.6** Discuss and debate the **impacts on jobs and work**
- 11.7** Discuss the arguments of **utopia and dystopia** in a debate of the future of robots and artificial intelligence (AI)
- 11.8** Understand the **data science** profession, specifically the role of a **citizen data scientist** in analytics

Opening Vignette (1 of 5)

Lessons Learned from Analytics Journey in an Organization

- The kind of problem you are solving with data science
 - Business problem versus research problem
- Defining the goal, value, and right business/research metrics
- Identify the data sources and check the data quality
 - Can we access the data that exist within our organization?
 - Have we considered data governance issues for data quality?

Opening Vignette (2 of 5)

Lessons Learned from Analytics Journey in an Organization

- **Prediction versus Causation/Explanation**
 - Bias Variance Tradeoff
 - Bias – how far off predictions are from true values
 - Variance – how much predictions fluctuate based on different training data
 - Overfitting versus Underfitting
 - Overfitting – a model learn training data too well, low training error but high test error
 - Underfitting – a model is too simple to learn, causing high error on both training and test data

Opening Vignette (3 of 5)

Lessons Learned from Analytics Journey in an Organization

- Maintenance of AI models
 - Retrain the models, when needed
- Creating the right framework for data science problems
 - Reports and dashboards
 - Exploratory data analysis, data visualization
 - Statistical analysis, hypothesis testing
 - Simple predictive models
 - Deep learning models
 - Causal models

Opening Vignette (4 of 5)

Lessons Learned from Analytics Journey in an Organization

- Explaining the model results to business and providing the recommendation
 - Explaining the results in terms of the business metrics
- Timeline expectations
 - Set realistic timeline expectations
- Involvement of business/product teams
 - Obtain business clients buy-in, frequently
- Talent gap
 - Finding and retaining good data scientists is a challenge

Opening Vignette (5 of 5)

Lessons Learned from Analytics Journey in an Organization

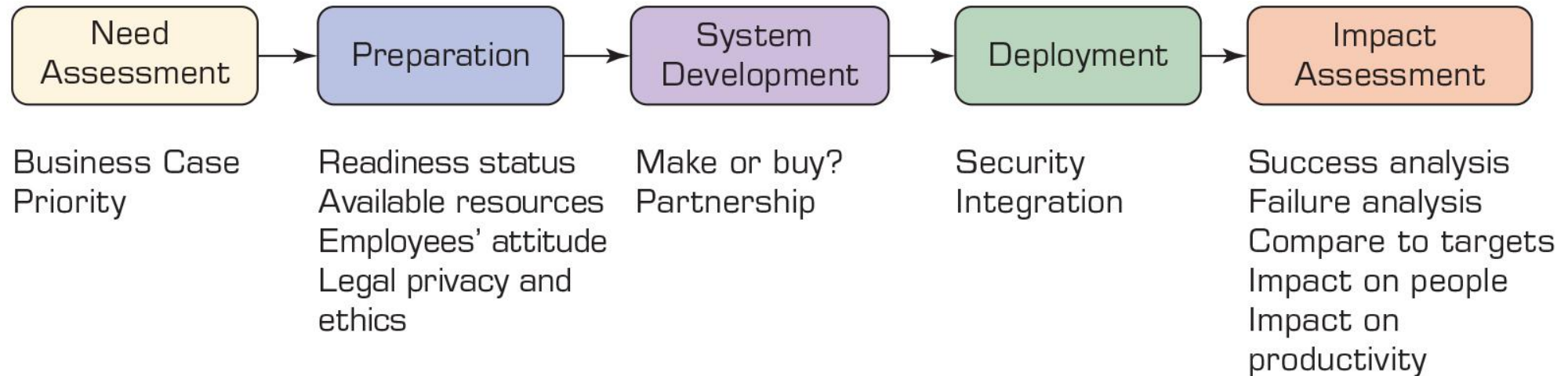
1. What is the difference between a business problem and a research problem?
2. Explain the difference between causation and predictions.
3. What are some of the data governance issues?

Implementing Intelligent Systems (1 of 2)

- Once the intelligent/AI system is developed, it needs to be put in use for the enterprise and knowledge workers
- Implementing business analytic/AI systems can be a complex undertaking
- Issues may be related to intelligent systems or computer-based information systems, in general
- You will learn about
 - The **process** to follow while implementing intelligent systems
 - The **challenges** and related issues to be aware of and properly deal with

Implementing Intelligent Systems (2 of 2)

- The Intelligent Systems Implementation Process
 - The implementation process of intelligent systems is similar to the generic implementation process of other information systems.



Successful Deployment of Intelligent Systems

- **Topics** related to deployment strategy include:
 - When to embark on intelligent projects and how to prioritize them.
 - How to decide whether to do it yourself or use partners, or to outsource.
 - How to justify investments in intelligent projects.
 - How to overcome employees' resistance (e.g., fear of job loss).
 - How to arrange appropriate people-robot teams.
 - How to determine which decisions to fully automate by AI.
 - How to protect intelligent systems (security) and how to protect privacy.
 - How to handle possible loss of jobs and retraining of employees
 - ...

Top Management and Implementation

- Senior executives need to
 - Understand the tactical as well as the strategic opportunities (of AI)
 - Redesign their organizations
 - Commit to helping shape the future of work.
- Not responding to the changes in a timely manner usually blamed on the senior managers
- Guidelines for digital labor (according to KPMG):
 - Establishing priority areas for technological innovation.
 - Developing a strategy and a plan for the employees.
 - Identify providers and partners for plans' execution.
 - Establishing a strategy and plans to realize benefits from the digital labor initiatives.”

System Development Implementation Issues (1 of 2)

- AI and business analytics maturity levels and related implementation issues may vary considerably
- **Three major problems** (according to Shchutskaya and Zharovskikh, 2020)
 1. **Development approach** – different from ordinary IT/computer systems development
 2. **Learning from data** – quality of the data determines the quality and usefulness of the outcomes
 3. **No clear view is available of how insights are generated** – as the stream of data generated change so may the insights
- Related important areas include problems with Big Data, ineffective information access, and limited integration capabilities

System Development Implementation Issues (2 of 2)

- **Connectivity and Integration**
 - As part of the development process, it is necessary to connect the AI and analytic applications to existing IT systems, including the Internet, and other intelligent systems.
 - Example: The Australian government commissioned Microsoft to build hyperscale cloud regions to unlock the power of intelligent technologies
- **Security Protection**
 - Many intelligent applications are managed and updated in the “cloud” and/or connected to the regular Internet; this may lead to new security vulnerabilities
 - Robots pose another source of danger and security threat

Leveraging Intelligent Systems in Business

- Leveraging the intelligent technology capabilities lead to increase efficiency and provide more customer care
 1. Customize the customer experience (e.g., for interactions with customers)
 2. Increase customer engagement (e.g., via chatbots)
 3. Use intelligent technologies to detect problems and anomalies in data.
- Critical success factors:
 - Discover, predict, justify, and learn from experience.
- Need to upgrade employees' skills and build an empowered AI-savvy workforce.
 - One of the most important issues: fear of job loss of employees.

Implementing IoT and Managerial Considerations (1 of 3)

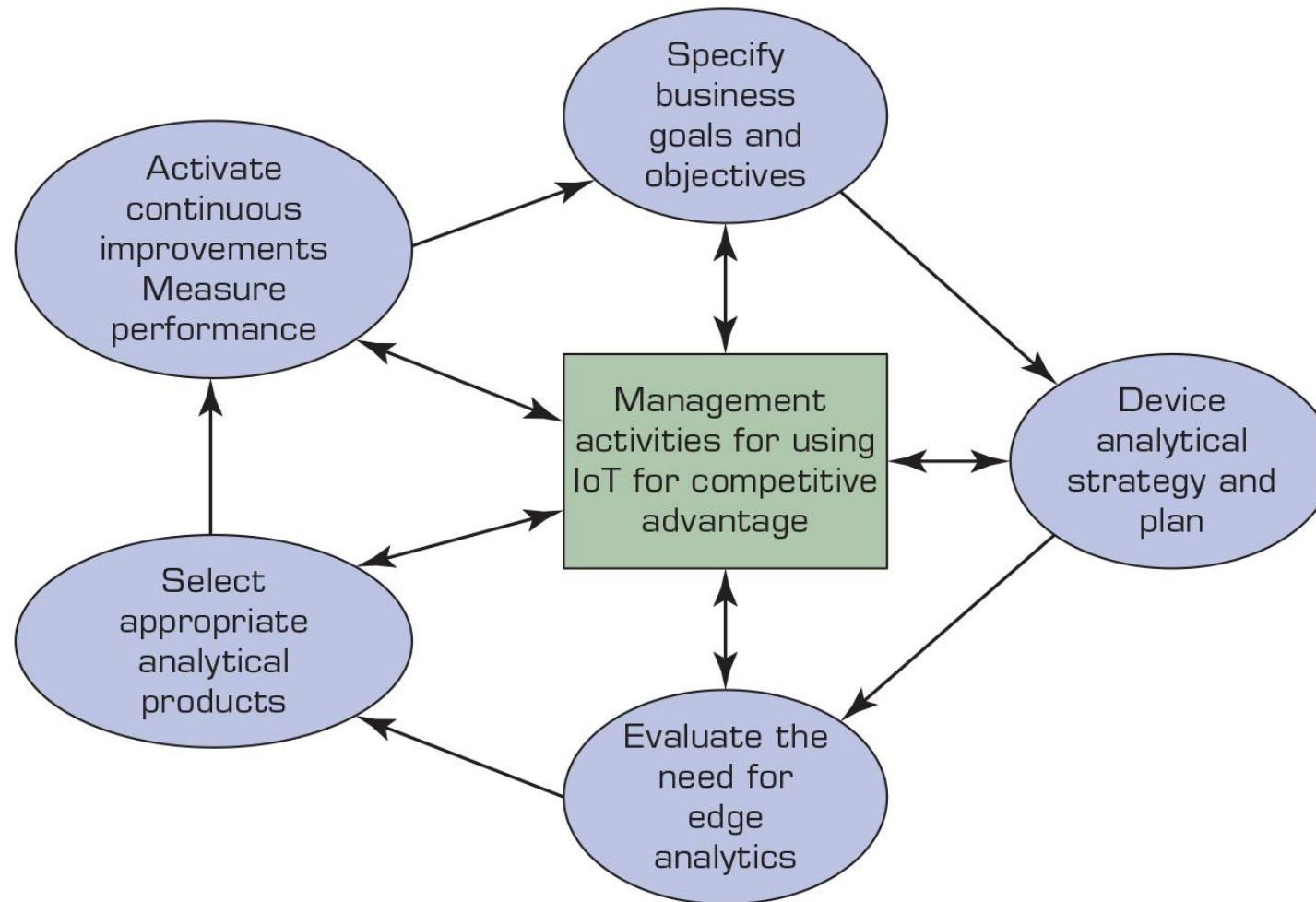
- **Major Implementation Issues:**
 - Organizational alignment
 - Interoperability challenges
 - Security
 - Privacy
 - Connection of the silos of data
 - Preparation of existing IT architectures and operating models for IoT can be a complex issue in many organizations
 - Management
 - Connected customers

Implementing IoT and Managerial Considerations (2 of 3)

- Strategy for Turning Industrial IoT into Competitive Advantage:
 1. Specify the business goals (including ROI analysis)
 2. Express an analytic strategy
 3. Evaluate the needs for **edge analytics** (analyzing data where is collected)
 4. Select appropriate analytics solutions
 5. Continues improvement closes the loop
- Stems for successful IoT implementation:
 - Develop a business case to justify the IoT project
 - Develop a working prototype
 - Install the IoT in one organizational unit as a pilot
 - Plan an organization-wide deployment

Implementing IoT and Managerial Considerations (3 of 3)

- The IoT Strategy Cycle (**Figure 11.2**)



Legal, Privacy and Ethical Issues (1 of 5)

- As data science, analytics, cognitive computing, and AI grow in reach and pervasiveness, everyone may be affected by these applications (positively or negatively)
- Just because something is doable through technology does not make it appropriate, legal, or ethical
- Data science and AI professionals and managers must be very aware of the related concerns
- Most legal, privacy, and ethical issues related to intelligent technologies are interconnected
 - For instance, many privacy issues are parts of ethics or have legal aspects

Legal, Privacy and Ethical Issues (2 of 5)

- **Legal issues** – many questions remain to be answered:
 - Who is liable for the actions of advice provided by intelligent machines?
 - Who is liable if an enterprise finds itself bankrupt as a result of using the advice of an AI-based application?
 - Will the enterprise itself be held responsible for not testing the system adequately before entrusting it with sensitive or volatile issues?
 - Will auditing and accounting firms share the liability for failing to apply adequate auditing tests?
 - Will the software developers of intelligent systems be jointly liable?
 - As self-driving cars become more common, who is liable for any damage or injury when a car's sensors, network, or AI system fail to function as planned?

Legal, Privacy and Ethical Issues (3 of 5)

- A Sample of **AI Potential Legal Issues**:
 - What is the value of an expert opinion in court when the expertise is encoded in a computer?
 - What happens if a physician accepts an incorrect diagnosis made by a computer and performs a procedure that results in the death of a patient?
 - What happens if a manager enters an incorrect judgment value into an intelligent application and the result is damage or a disaster?
 - Can management force experts to contribute their expertise to an intelligent system? How will they be compensated?
 - Who owns the knowledge that the new AI systems such as Chat GPT and Google's Bard gather and is then employed by an end user in some application?
 - ...

Legal, Privacy and Ethical Issues (4 of 5)

- **Legal Issues**

- In the case of intelligent technologies, Intellectual Property Protection issues present some unique dimensions
- Several of the ethical issues need to be combined with legal issues
 - For example, take robots' legal rights. Do we need these rights? What for (an ethical issue)?
 - Facebook, for example, has had legal issues regarding face recognition
 - Safety rules for robots were developed a long time ago
- At the moment, there are very few laws regarding intelligent technologies. Most of the laws relate to safety.

Legal, Privacy and Ethical Issues (5 of 5)

- **AI in Law** – AI applications to the legal profession and the solution of some legal problems:
 - Analyzing legal-related data (e.g., regulatory conflicts) to detect pattern
 - Providing legal advice to consumers
 - Document review
 - Analyzing contracts
 - Supporting legal research
 - Predicting results (e.g., likelihood to win)
 - AI impact on the legal profession
- Ownership of content used for training AI models
 - **Interesting read:**
<https://news.bloomberglaw.com/ip-law/first-ai-art-generator-lawsuits-threaten-future-of-emerging-tech>

Privacy Issues (1 of 7)

- Privacy means different things to different people.
- **Privacy** is the right to be left alone and the right to be free from unreasonable personal intrusions
- Privacy has long been related to legal, ethical, and social issues in many countries
- The right to privacy is recognized in every state of the US
- Two rules have been followed in past court decisions:
 - The right of privacy is not absolute. Privacy must be balanced against the needs of society
 - The public's right to know is superior to the individual's right to privacy

Privacy Issues (2 of 7)

- Collecting information about individuals
 - Intelligent technologies aim to provide targeted services and marketing to consumers by collecting information about these customers
 - The Internet with large-scale databases has created an entirely new dimension of accessing and using data
- Data from internet can also be used for **the good of society**
 - Examples include:
 - Eliminate or reduce fraud, prevent crime
 - Identify and prevent government mismanagement, tax evasion, welfare cheating, family-support filching, employment of illegal workers, and so on.

Privacy Issues (3 of 7)

- **Virtual Personal Assistants**
 - Amazon's Echo/Alexa, Google Assistant, iPhone Siri, and other similar devices listen to what is going on
 - Your voice assistant is spying on you!
 - Alexa can act as a fashion consultant using *style check*
 - Privacy versus convenience
- **Mobile User Privacy**
 - Many users are unaware of the private information being tracked through their smartphone usage
 - Many apps collect user data that track from
 - Phone call as it moves between cell towers
 - GPS-enabled devices that transmit users' locations
 - Phones transmitting information at Wi-Fi hotspots

Privacy Issues (4 of 7)

- **Privacy in IoT Networks**
 - More data are flowing with IoT networks
 - How do we protect the data and guard against their misuse?
- **Recent technology issues in privacy and analytics**
 - Companies have started to employ intelligent technologies to develop profiles of users on the basis of their device usage, surfing, and contacts
 - “What They Know” (WallStreetJournal.com, 2016)
 - Rapleaf (now part of Towerdata) – from email address to complete profile creations [privacy and ethical issue?]
 - BlueCava technology – multi device personal profile detection
 - Many of the analytics start-ups in this space claim to honor user privacy, but violations are often reported
 - Reputation. Com – using just an email address to find SSN!

Privacy Issues (5 of 7)

- **At work:** analyzing employee behaviors based on the data collected from sensors that employees wear in a badge
 - Humanyze - reported several such applications of its sensor-embedded badges
 - These sensors track all movements of an employee.
 - Given rise to another term: **workplace analytics**
- **Example:** Using sensors and IoT to observe bankers at Barclays bank (London, United Kingdom)
 - Using heat and motion sensors, Barclays tracks how long its bankers are at their desks.
 - They said – they did it to optimally allocate office space
 - The union is watching this IoT application carefully to ensure that it is not used to spy on employees

Privacy Issues (6 of 7)

- Other Issues of Potential Privacy Violation
- The following are some more examples of **potential privacy violations** in the intelligent technology world:
 - Delaware police are using AI dashcams to look for fugitives in passing cars. Photos and videos taken are sent to the clouds and analyzed there by AI algorithms
 - Facebook's face recognition systems create concerns regarding privacy protection
 - Epicenter offers its employees a microchip implant. It acts like a swipe card, opens doors, buys you food in the company store, and much more. But management can track you too. It is given only to volunteers

Privacy Issues (7 of 7)

30

- **Who Owns Your Data?**

- New technologies allow easier data capture and mine
- New cars comes with many sensors, IoT, and GPS devices
 - A car connected to the Internet → A privacy nightmare
- Apple (CarPlay) and Google (Android Auto) – who own this data, who can have access these data?
- Many other apps like Google Waze, Yelp, Spotify popularly used in cars collect all kinds of data
- The bottom line - Intelligent systems professionals and users must be aware of the legal and ethical issues involved in collecting information that may be privileged or protected
- Privacy issues are considered in many cases as important components of ethics.

Ethical Issues (1 of 5)

- Personal values constitute a major factor in the issue of ethical decision-making
- The study of ethical issues is complex because of their multidimensional nature
- **Some ethical issues are upsetting, but they are not illegal**
 - Example: Facebook's experiment to present different News Feeds to the users and monitor their emotional reactions as measured by replies, likes, sentiment analysis, and so on.
 - Many other companies are doing similar analytics to improve their product or service offerings
 - Because Facebook is so large, running this experiment without the users' informed consent was viewed as unethical
 - Facebook acknowledged its error and instituted a formal review

Ethical Issues (2 of 5)

- Questions regarding the **ethical issues** in AI, robotics, and other intelligent systems
 - What is their impact on jobs?
 - How do machines (i.e., robots) affect our behavior and interactions?
 - How can wealth created by intelligent machines be distributed?
 - How can intelligent applications mistakes be guarded against?
 - Can intelligent systems be fair and unbiased? How can bias in creation and operation of AI systems be eliminated?
 - How can intelligent applications be kept safe from adversaries?
 - How can systems be protected against unintended consequences (e.g., accidents in robot operations)?
 - How can we stay in control of a complex intelligent system?

Ethical Issues (3 of 5)

- **Additional issues** related to ethical intelligent systems
 - Electronic surveillance
 - Ethics in business intelligence (BI) and AI systems design
 - Software piracy
 - Invasion of individuals' privacy
 - Use of proprietary databases and knowledge bases
 - Use of personal intellectual property for the benefits of companies and the payment to the contributors
 - Accuracy of data, information, and knowledge
 - Protection of the rights of users
 - Accessibility to information by AI users
 - The amount of decision-making to delegate to intelligent machines.
 - How AI can fail due to inappropriate ethics

Ethical Issues (4 of 5)

- Other Topics in Intelligent Systems Ethics
 - Machine ethics is a part of the ethics of AI that is concerned with the **moral behavior** of artificially intelligent beings
 - Robotics is concerned with the moral behavior of designers, builders, and users of robots
 - Microsoft's Tay chatbot was closed due to its inability to understand many irrelevant and offending comments
 - Some are afraid that algorithm-based technologies, including AI, may become racists
 - **Self-driving cars** may one day face a decision of whom to save and whom to kill
 - Voice technologies enable the identification of callers to AI machines.
 - Ethical issues in healthcare and medical field

Ethical Issues (5 of 5)

- **The 10 Commandments of Computer Ethics**

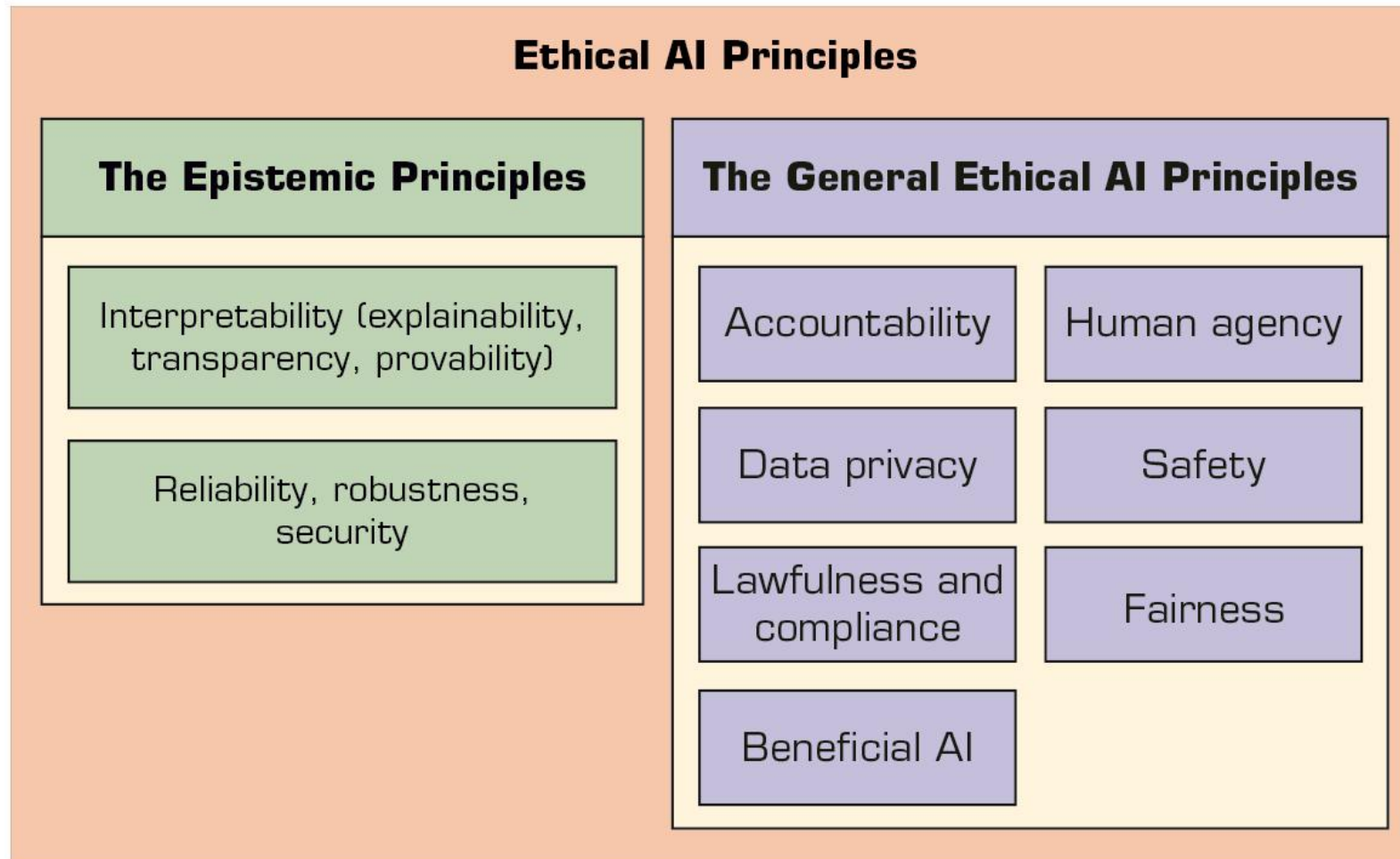
1. Thou shalt not use a computer to harm other people.
2. Thou shalt not interfere with other people's computer work.
3. Thou shalt not snoop around in other people's files.
4. Thou shalt not use a computer to steal.
5. Thou shalt not use a computer to bear false witness.
6. Thou shalt not use or copy software for which you have not paid.
7. Thou shalt not use other people's computer resources without authorization.
8. Thou shalt not appropriate other people's intellectual output.
9. Thou shalt not think about the social consequences of the program you write.
10. Thou shalt not use a computer in ways that show consideration and respect.

Ethical/Responsible/Trustworthy AI (1 of 4)

- Related stories:
 - The Curious Case of BlenderBot 3.0
 - Amazon's Scrapped AI-Powered Recruiting Tool
 - Correctional Offender Management Profiling for Alternative Sanctions (COMPAS)
- Principles of Responsible AI
 1. Fairness
 2. Reliability and safety
 3. Privacy and security
 4. Inclusiveness
 5. Transparency
 6. Accountability

Ethical/Responsible/Trustworthy AI (2 of 4)

- Ethical Principles of AI (by Dr. Anand Rao at PwC)



Ethical/Responsible/Trustworthy AI (3 of 4)

- Why Ethical AI Should Matter to the Managers?
 - A successful implementation of AI technologies: **Netflix**
 - DVD sales to rentals to streaming to production to ..
 - **Recommender systems** are built using AI algorithms and a lot of user behavioral data
 - These systems are not immune to biases
 - Fortune Business Insights: The global AI market is expected to grow from \$387.45 billion in 2022 to \$1,394.30 billion by 2029
 - Accenture: AI has the potential to increase profitability rates by an average of 38% by 2035
 - **In summary**: AI is a dominant player in the technology industry, and it is going to become an indispensable tool for even more organizations

Ethical/Responsible/Trustworthy AI (4 of 4)

- The O'Neil **Claim of Potential Analytics' Dangers**
 - Managers and data scientists should be aware of the social and long-term effects of mathematical models and algorithms
 - *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy* (a popular book O'Neil)
 - O'Neil: **Models** must satisfy **three** conditions
 1. They must be transparent. That is, if the model is not understandable, its application can lead to unintended consequences
 2. The model must have clear quantifiable objectives
 3. The third requirement is that the models must have a self-correcting mechanism and a process in place so that they are audited regularly, and new inputs and outputs are constantly being considered

Impacts of Intelligent Systems on Organizations (1 of 2)

- Explosive growth in analytics, AI, and cognitive computing is going to have a major impact on the future of organizations
- **New organizational units** and their management
 - Creating a new analytics department, a BI department, a data science department, and/or an AI department
 - C-suite titles - Chief Data Officer, Chief AI officer, and so on
- Transforming businesses and increasing competitive advantage
 - Enhancing the **digital transformation**
- Using intelligent systems to gain competitive advantage
 - Amazon uses robots to reduce cost and control online commerce

Application Case 11.1

How 1-800-Flowers.com Uses Intelligent Systems for Competitive Advantage

Questions for Discussion

1. Why it is necessary to provide better customer experience today?
2. Why do data need sophisticated analytical tools?
3. Read the “Key benefit of SAS Marketing Automation.” Which benefits do you think are used by 1-800-Flowers.com and why?
4. Relate IBM Watson to “personalization.”
5. Relate “SAS Advanced Analytics” capabilities to their use in this case.
6. “SAS Enterprise Miner” is used to do data mining. Explain what is done and how.

Impacts of Intelligent Systems on Organizations (2 of 2)

- **Intelligent Systems' Impact** on Managers' Activities, Performance, and Job Satisfaction
 - Less expertise (experience) is required for making many decisions.
 - Faster decision-making
 - Less reliance on experts and analysts
 - Power is being redistributed among managers - the more information and analysis capability they possess, the more power they have
 - Support for complex decisions
 - Information needed for high-level decision-making is expedited or even self-generated
 - Automation of routine decisions or phases in the decision-making process

Impacts on Job and Work (1 of 8)

- One of the most discussed and debated topics in considering the impacts of intelligent systems is on **jobs and work**. There is a general **agreement** that:
 - Intelligent systems will create many new jobs as automation always has
 - There will be a need to retrain many people
 - The nature of work will be changed
- There is no consensus among the experts on when, how much, and how to deal with this phenomena
- AI, will accelerate the *polarization* of the labor market
 - Growth in the top and bottom tiers of the labor market but losses in the middle

Impacts on Job and Work (2 of 8)

- **Are Intelligent Systems Going to Take Jobs—My Job?**

- Tesla's Elon Musk envisions AI-based autonomous driving trucks all over the world within 10 years.
 - There will be convoys of such trucks, each of which will follow a lead truck.
 - Trucks will be electrical, economical, and pollution free. In addition, there will be fewer accidents.
 - But what about thousands of drivers who will lose their jobs?
- Amazon has opened its first Go, a cashierless physical store. They plan 3,000 more in a few years.
- The post office in some countries already distributes mail using autonomous vehicles.
- In short, there is a chance for massive unemployment.

Impacts on Job and Work (3 of 8)

- Intelligent Systems May Create **Massive Job Losses**
 - They are moving very fast
 - They may take a large variety of jobs, including many office-focused, administrative, and nonphysical jobs.
 - Their **comparative advantage over manual labor** is very large and growing rapidly.
 - They are already taking some professional jobs from financial advisors, paralegals, and medical specialists.
 - The capabilities of AI are growing rapidly.
 - In Russia, robots are already teaching mathematics in schools (some do a better job than humans). Just think about what could happen to the teaching profession..

Application Case 11.2

Administrative/Office Jobs That Robots Have Already Taken

- **FedEx** will have pilotless airplanes in the near future, and schools will have no human teachers!
- **Jobs already taken by robots:**
 - Sherman (2015): Online marketers, financial analysts and advisors, anesthesiologists, diagnosticians, and surgeons, financial and sports reporters
 - Palmer (2017): middle management, commodity salespeople, report writers, accountants and bookkeepers, and some types of doctors.
 - McFarland (2017b): cashiers, toll booth operators, fast-food employees, and drivers. Low-risk jobs include nurses, doctors, dentists, youth sport coaches, and social workers.

Impacts on Job and Work (4 of 8)

- Probability of Job Loss

Low-Risk Jobs

0.0036 First-Line supervisors of firefighting and prevention workers

0.0036 Oral and maxillofacial surgeons

0.0035 Healthcare social workers

0.0035 Orthotists and prosthetists

0.0033 Audiologists

0.0031 Mental health and substance abuse social workers

0.0030 Emergency management directors

0.0030 First-line supervisors of mechanics, installers, and repairers

0.0028 Recreational therapists

Impacts on Job and Work (5 of 8)

- Probability of Job Loss

High-Risk Jobs

0.99 Telemarketers

0.99 Title examiners, abstractors, and searchers

0.99 Sewers, hand

0.99 Mathematical technicians

0.99 Insurance underwriters

0.99 Watch repairer

0.99 Cargo and freight agents

0.99 Tax preparers

0.99 Photographic process workers and processing machine operators

0.99 New account clerks

Impacts on Job and Work (6 of 8)

- Jobs and the Nature of Work Will Change
- Low-skill jobs will be taken by machines, but **high-skill jobs may not**. Jobs may be redesigned either
 - to be low skilled in order to be automated
 - to be high skilled so that they will be executed by humans.
- Manyika of McKinsey & Company
 - Many activities done by humans will have the potential to be automated.
 - Productivity growth from robotics, AI, and machine learning will be tripled
 - AI will create many new jobs paying high salaries.
 - Since more than half the world is still offline, the changes will not be too rapid.

Impacts on Job and Work (7 of 8)

- **Tips for Success**

- Digital capabilities need to come before AI.
- Machine learning is powerful, but it is not the solution to all problems.
- Do not put technology teams solely in charge of intelligent technologies.
- Adding a business partner may help with AI-based projects.
- Prioritize a portfolio approach to AI initiatives.
- The biggest challenges will be people and business processes.
- Not every business is using intelligent systems, but almost all those that use them increase income and profit.
- Top leadership support is necessary for a transformation to AI.

Impacts on Job and Work (8 of 8)

- Dealing with the changes in jobs and the nature of work
 1. Use learning and education to facilitate the change.
 2. Involve the private sector in enhancing training and retraining.
 3. Have governments provide incentives to the private sector so employees can invest in improved human capital.
 4. Encourage private and public sectors to create appropriate digital infrastructure.
 5. Innovative income and wage schemes need to be developed.
 6. Carefully plan the transition to the new work. Deal properly with displaced employees.
 7. Properly handle new technology-enabled technologies.
 8. Focus on new job creation, particularly digital jobs.
 9. Properly capture the productivity increase opportunities.

What are potential dangers
of AI?

Potential Dangers of AI (1 of 2)

- Elon Musk:
 - We need to be super careful with AI. Potentially more dangerous than nukes.” (See or listen to the video at youtube.com/watch?v=S YqCbJ0AqR4) Musk predicts that World War III will start because of AI. “Robots will kill us all, one day,” he said in his several presentations.
- Bill Gates:
 - “I am in the camp that is concerned about super intelligence. Musk and some others are on this, and I don’t understand why some people are not concerned.”
- Stephen Hawking:
 - The late scientist stated, “The development of full artificial intelligence could spell the end of the human race.”

Potential Dangers of AI (2 of 2)

- The AI Utopia's Position
 - Future of AI: watch at youtube.com/watch?v=UzT3Tkwx17A
 - AI to the quality of life - 20% reduction in crime
- Some issues related to the utopia:
 1. AI will be so great that people will have a problem of what to do with their free time.
 2. The road to AI Utopia could be rocky, for example, there will be impacts on jobs and work.
 3. One day we will not drive anymore and there may not be human financial advisors; everything will be different, and the changes may be rapid and turbulent, and we may even face disasters, as projected by the Dystopia camp.

Citizen Science & Citizen Data Scientist

- **Citizen science:** members of general public voluntarily help conduct scientific research
 - At any stage of the scientific process—design experiments, collect data, help analyze data, develop reports, etc.
- **Citizen data scientist:** a business user in an organization who has not been fully trained in mathematics and statistics concepts of data science but has learned to use the software tools to build models for their own business use.
 - Use of analytics tools that are visual and intuitive
 - Help alleviate the shortage of data science professionals
- New topics are constantly coming up in the field of analytics/data science/AI. Some are likely to be just fads whereas the others will make transformative difference.